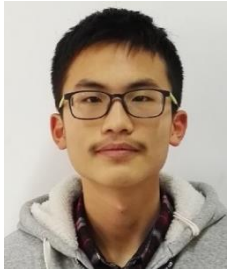
CV



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Shibiao Jiang

Zhejiang University

Computer Science and Technology

EDUCATION

Zhejiang University B.S. Computer Science and Technology 09/2017 - 06/2021

Study in Qiushi Honor's Program of Chu Kochen Honors College

GPA: 4.41/5.00 (3.94/4.00) Rank: 1/20 2017-2018

4.46/5.00 (3.92/4.00) Rank: 5/20 2018-2019

Programming Language and Skills: C | C++ | Python(Pytorch) | Java | Latex | Verilog | SQL

Relevant Courses: Linear Algebra, Mathematical Analysis, Object-Oriented Prigramming, Python Programming, Discrete Mathematics, Probability Theory, Advanced Data Structure, Operating System, Java Applied Technology, Operating System, Computer Vision, Theory of Computation, Computer Network, Introduction to Data Mining……

SCIENTIFIC EXPERIENCE

1. Researcher in Tencent(Shanghai) YouTu Lab 07/2019 - 08/2019

Tencent Youtu Lab is an enterprise laboratory focusing on machine learning. *Mentor: Chao Chen*

- ✧ I Joined the Fundamental Research Group II in Youtu Lab. My mentor is Chao Chen, who is the leader of Faces Archiving Project, and I assisted him in improving the accuracy of face clustering.
- ✧ I **proposed a new face clustering model using Bayes Analysis**. In the usual case, the graph of faces will be built by eigenvectors which are generated through complex layered convolutional networks, and traditional clustering methods like k-means or Rank-Order will be directly applied. However, I figured out that particular methods could be applied for special situations. Then I derived a series of formulae using Bayes Analysis to correct the similarity between faces with the help of prior probabilities, which can be calculated in advance from big data.
- ✧ My new approach has **greatly improved the accuracy of face clustering**. In my last few days of internship, I was involved in engineering this method and test its effect in real situation, and found that **it even beat the mature system developed by YouTu Lab**.

2. Joined the Laboratory of Visual Intelligence and Pattern Analysis at Zhejiang University 09/2019 - now

Introduction for VIPA: https://person.zju.edu.cn/msong#570836 *Supervisor: Mingli Song*

- ✧ I joined ZJU-VIPA Lab in my junior year. Professor Mingli Song is devoted to make deep learning interpretable in computer vision, and I joined one of his groups which focuses on Document Analysis. The first stage of this project is to extract features from the text scenes and comprehend the meaning of words.
- ✧ To achieve the goal, we're researching in some special cases now, like **restoring invoices** which may be creased or folded. The knowledge of both computer vision and natural language processing are involved in this case.
- ✧ We tried many SOTA models and successfully built a complete pipeline for invoices restoring, and proposed some new methods. Finally our accuracy approaches the result supported by big company.

3. Remote Research in UCLA-VCLA Group 03/2020 - 05/2020

Introduction for VIPA: https://person.zju.edu.cn/msong#570836 *Supervisor: Songchun Zhu*

- ✧ I was admitted in VLCA (Vision, Cognition, Learning, and Autonomy) Lab at the University of California, Los Angeles as a summer intern, under the instruction of Professor Songchun Zhu and Ph.d. Pan Lu.
- ✧ Due to the terrible situation of COVID-19, we decided to do remote research in advance and now we are preparing for the EMNLP2020 (due in mid-May).

HONORS & AWARDS

Silver Medal in 2016 National Olympiad of Informatics	National
First Prize for both 2017 and 2018 **Fundamental Research Scholarship** of Qiushi Honor's Program	Zhejiang University
Second Prize for both 2017 and 2018 **Zhejiang University Scholarship**	Zhejiang University
Scholarship of **Zhejiang Provincial Government**	Zhejiang Province
First Prize for 2017 42nd International Collegiate Programming Contest Beijing Onsite	International
First Prize for 2017 42nd International Collegiate Programming Contest East-Continent League Finals	International
Runner-up for 2017 2nd China Collegiate Programming Contest Finals	National
Champion for 2018 19th Zhejiang University Programming Contest	Zhejiang University
First Prize for 2018 43rd International Collegiate Programming Contest East-Continent League Finals	International
Second Runner-up for 2018 43rd International Collegiate Programming Contest Beijing Onsite	International
Second Runner-up for 2018 43rd International Collegiate Programming Contest Nanjing Onsite	International
Rank 21st for 2019 43rd International Collegiate Programming Contest **World Finals**	International
First Prize for 2019 Huawei Jingying Scholarship	Huawei Company

SELECTED PROJECTS

Mua-Interpreter: Principle of Programming Language

https://github.com/jiangshibiao/MUA-Interpreter *Instructor: Prof. Kai Weng, Zhejiang University*

- ✧ Mua is a specially designed functional programming language created by my professor Kai Weng. It supports list operations, variable bindings and function definitions, and I implemented an interpreter for Mua using Java.
- ✧ The interface is quite like Python that the interpreter will quickly respond to the user's real-time commands.
- ✧ This project involves many advanced idea, like the reflection mechanism in Java, the object-oriented programming combined with the functional programming, and the thought of stream computing.

PumpkinBattle: Computer Graphics

https://github.com/jiangshibiao/pumpkinBattle *Instructor:Prof. Ruofeng Tong, Zhejiang University*

- ✧ I Designed and implemented an interesting game using OpenGL in C++.
- ✧ Surrounded with walls in the dark environment, the player is supposed to walk and jump in the scene without being caught by pumpkin monsters which are wandering. And the player can use the explosive to blow up some pieces of the wall to get through it, or use a gun to shoot monsters.
- ✧ Involved many CG technologies such as illumination models, real-time shadow, collision detection, etc.

MINISQL: Database System

https://github.com/jiangshibiao/MiniSQL *Instructor: Prof. Yunjun Gao, Zhejiang University*

- ✧ I Designed a simple SQL engine in C++ which supports SQL-like commands and gives real-time responses.
- ✧ Through the careful implementation of B + tree and buffer management, this engine runs very fast.

You can find more projects on https://github.com/jiangshibiao/Course-Review.

OVERSEAS EXPERIENCE

Attendee for Moscow Pre-Finals Workshop 08/2018

- ✧ Invited by the Moscow Institute of Physics and Technology, I have been to Russia for 12 days (8.20-8.31) to attend Moscow Workshop. During the time I communicated and competed with the top students from all over the world.

Participant for 2019 ICPC World Finals in Portugal 04/2019

- ✧ On behalf of my university, I was invited by University of Porto and took part in the 2019 ICPC World Finals in Porto, Portugal. There are 135 top teams from all over the world came to this competition.
- ✧ Finally I won the 21st place, which is the best result for Zhejiang University in recent four years.

Problem Setter and Reviewer for ICPC2019 and CCPC2019 09/2019 - 11/2019

- ✧ I was the problem setter and reviewer for ICPC2019 Hong Kong Onsite and CCPC2019 Harbin Onsite.
- ✧ Besides designing and reviewing problems, I was also the referee and the technical director during the onsite contest.